

2023 Bi H1 Q15

Section: Metabolism and Survival

Topic: Metabolism and Adverse Conditions

Question Summary

This question investigates an experiment testing how sunscreen affects the survival of yeast cells exposed to ultraviolet (UV) light. You must choose which control experiment would allow a valid conclusion about the sunscreen's effectiveness.

Worked Solution

The experiment's aim is to find out whether sunscreen increases the survival of yeast under UV light. The test setup involves yeast on agar, with sunscreen spread on the lid, then exposure to UV light. To draw a valid conclusion, the only difference between the test and control must be the presence or absence of sunscreen.

Therefore, the control should contain:

- Yeast (so it is comparable to the test)
- No sunscreen, since that's the factor being tested
- UV exposure kept the same

■ Answer: B — Sunscreen

Revision Tips

- A control experiment keeps all conditions identical except for one variable — the one being tested.
- This allows the effect of that variable (in this case, sunscreen) to be isolated and compared.
- Biological controls are essential for reliable conclusions.
- Example: testing antibiotics, disinfectants, or temperature tolerance — always remove one variable at a time.